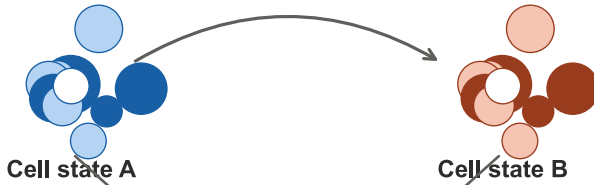


CKI: A Cell-state Kinetic Index

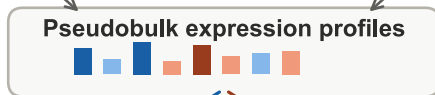
Quantifying functional cell-type divergence, normalized to the housekeeping baseline

compare expression profiles

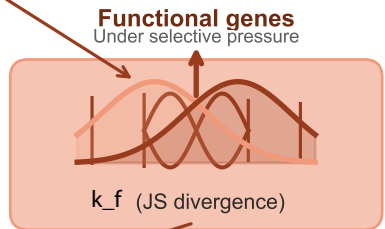
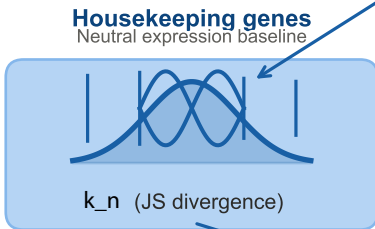
1



2



3



4

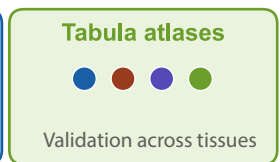
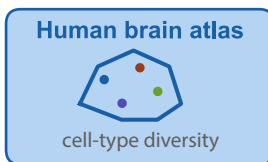
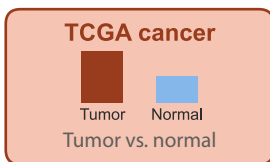
$$\omega = k_f / k_n$$

$\omega > 1 \rightarrow$ divergence exceeds neutral baseline

5



Applications



Key insight:

ω distinguishes functional remodeling from neutral transcriptomic drift